

Diagonalization

Observation :

$$A = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{bmatrix} \in M_{n \times n}(F)$$

"Diagonal matrix"

all non-zero entries are on the diagonal

Then

$$\underbrace{A \cdot A \cdot \dots \cdot A}_{m\text{-times iterated product}} = A^m \quad \text{is very easy to compute}$$

In fact if

$$A = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}, \quad B = \begin{bmatrix} \mu_1 & & 0 \\ & \ddots & \\ 0 & & \mu_n \end{bmatrix}$$

are two diagonal matrices in

$M_{nm}(F)$ then

$$AB = \begin{bmatrix} \lambda_1 \mu_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \mu_n \end{bmatrix}$$

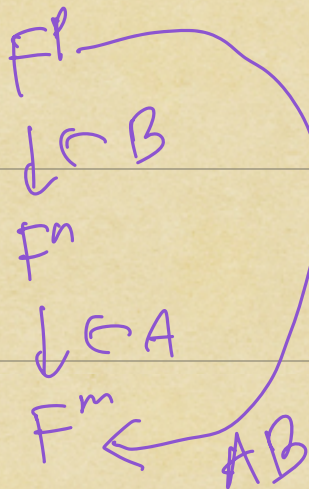
(= BA)
very special
to diagonal
matrices

$$A^n = \begin{bmatrix} \lambda_1^n & & 0 \\ & \ddots & \\ 0 & & \lambda_n^n \end{bmatrix}$$

Matrix multiplication

$$\underbrace{A}_{m \times n} \underbrace{B}_{n \times p} = A \underbrace{[v_1 \ v_2 \ \dots \ v_p]}_{(n \times 1) \text{ - column vectors}}$$

$$= [Av_1 \ Av_2 \ \dots \ Av_p]$$



To compute

$$Av \quad \text{for} \quad v = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \in F^n$$

note that Ae_i is the i -th column of A

Since $v = c_1 e_1 + \dots + c_n e_n$

$$\begin{aligned} \Rightarrow Av &= c_1 \cdot (\text{First column of } A) + \dots + c_n (\text{n-th column of } A) \\ &= c_1 \begin{bmatrix} a_{1,1} \\ a_{2,1} \end{bmatrix} + \dots + c_n \begin{bmatrix} a_{1,n} \\ a_{2,n} \end{bmatrix} \end{aligned}$$

$$\begin{bmatrix} \vdots \\ a_{m,1} \end{bmatrix}$$

$$\begin{bmatrix} \vdots \\ a_{m,n} \end{bmatrix}$$

$$= \begin{bmatrix} a_{1,1}c_1 + a_{1,2}c_2 + \dots + a_{1,n}c_n \\ a_{2,1}c_1 + a_{2,2}c_2 + \dots + a_{2,n}c_n \\ \vdots \\ a_{m,1}c_1 + a_{m,2}c_2 + \dots + a_{m,n}c_n \end{bmatrix}$$

This can be interpreted as follows

$$\begin{bmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \dots & a_{m,n} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \rightarrow a_{1,1}c_1 + \dots + a_{1,n}c_n$$

||

$$\begin{bmatrix} a_{1,1} \\ a_{1,2} \\ \vdots \\ a_{1,n} \end{bmatrix} \cdot \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

Defⁿ The dot product

between two vectors

$V, W \in F^n$ is the scalar

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \quad \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$$V \cdot W = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

Remark: The dot product is actually a special case of the matrix product!

$$V = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in F^n \rightsquigarrow V^T = [a_1 \dots a_n]$$

↑
"The transpose of V "

$$[V \cdot W] = \underline{V}^T \underline{W}$$

$$(\begin{smallmatrix} n & n \end{smallmatrix})$$

$[n] - \text{matrices}$

So another way of writing matrix products

$$A = \begin{bmatrix} w_1^T \\ w_2^T \\ \vdots \\ w_m^T \end{bmatrix}, \quad B = [v_1 \ v_2 \ \dots \ v_p]$$

$$w_1, \dots, w_m \in F^n$$

$$\underbrace{AB}_{m \times p} = \begin{bmatrix} w_1 \cdot v_1 & w_1 \cdot v_2 & \dots & w_1 \cdot v_p \\ w_2 \cdot v_1 & \dots & \dots & w_2 \cdot v_p \\ \vdots & & & \vdots \\ w_m \cdot v_1 & \dots & \dots & w_m \cdot v_p \end{bmatrix}$$

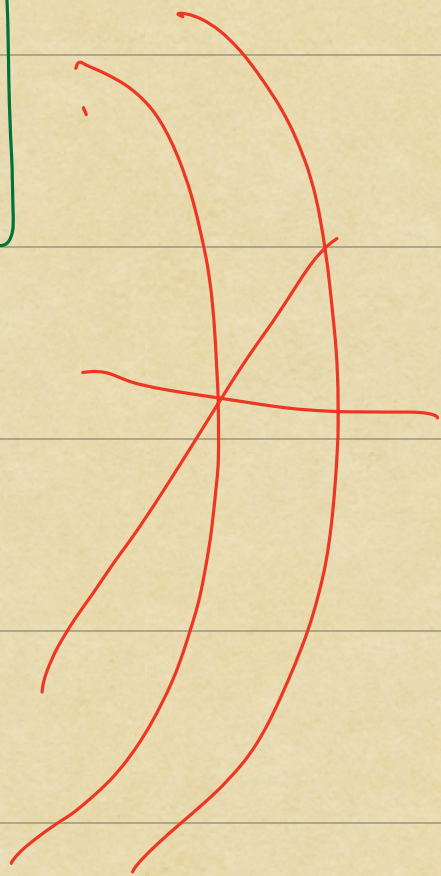
Example

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 3 \\ 21 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 6 \\ 9 & 11 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}^3 = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 4 \\ 8 & 11 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 11 & 15 \\ 30 & 41 \end{bmatrix}$$

Observation:

$A \in M_{n \times n}(F)$ corresponding
to $T: F^n \longrightarrow F^n$

Then A is diagonal, $A = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$

$$(\Leftrightarrow) T(e_i) = \lambda_i e_i$$

Defⁿ

V : finite dim'l F -vector space

$$T: V \longrightarrow V$$

is diagonalizable if the following equivalent conditions hold:

(1) $\exists$ a basis $B = (v_1, v_2, \dots, v_n)$ for V and scalars $\lambda_1, \dots, \lambda_n \in F$ s.t.

$$T(v_i) = \lambda_i v_i, \quad i = 1, \dots, n$$

(2) $\exists$ a basis $B = (v_1, v_2, \dots, v_n)$

s.t.

$[T]_B$ is diagonal

$$\uparrow \\ M_{n \times n}(F)$$

